

Support Vector Machine (SVM) - Complete Notes

1. Introduction

Support Vector Machine (SVM) ek **Supervised Machine Learning Algorithm** hai jo mainly **Classification** ke liye use hota hai. Ye Regression (SVR) ke liye bhi use ho sakta hai.

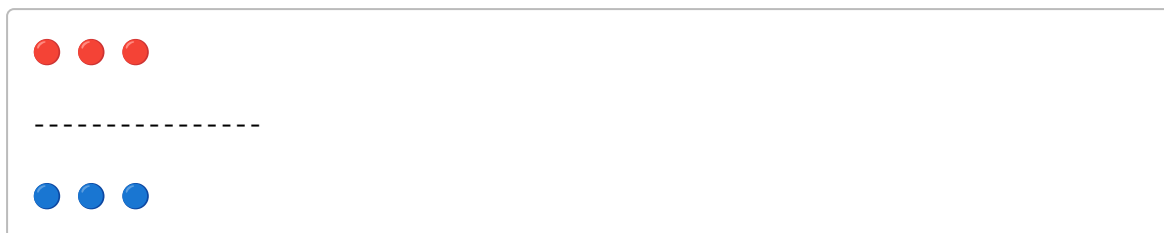
SVM ka main objective hai:

Aisi Decision Boundary (Hyperplane) find karna jo do classes ke beech maximum margin create kare.

2. Decision Boundary

Decision Boundary wo line (2D), plane (3D) ya hyperplane (n-D) hoti hai jo different classes ko separate karti hai.

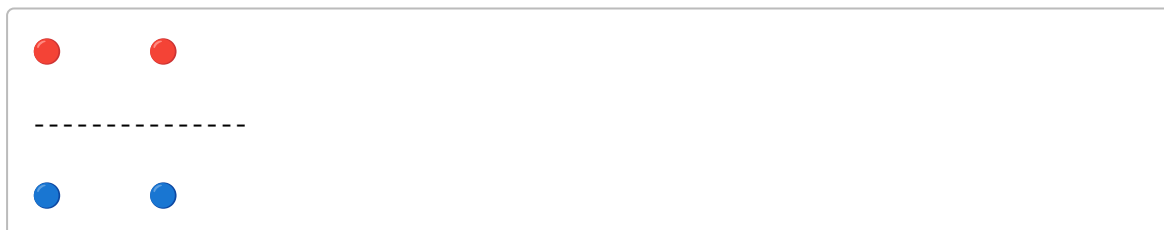
Example:



Bahut si lines classes ko separate kar sakti hain, lekin SVM unme se **best line** choose karta hai.

3. Margin

Margin = Decision Boundary aur nearest data points ke beech ki distance.



Rule

- Margin jitna bada hoga
- Model utna robust hoga

- Generalization utni better hogi

Isliye SVM ko **Maximum Margin Classifier** bhi kehte hain.

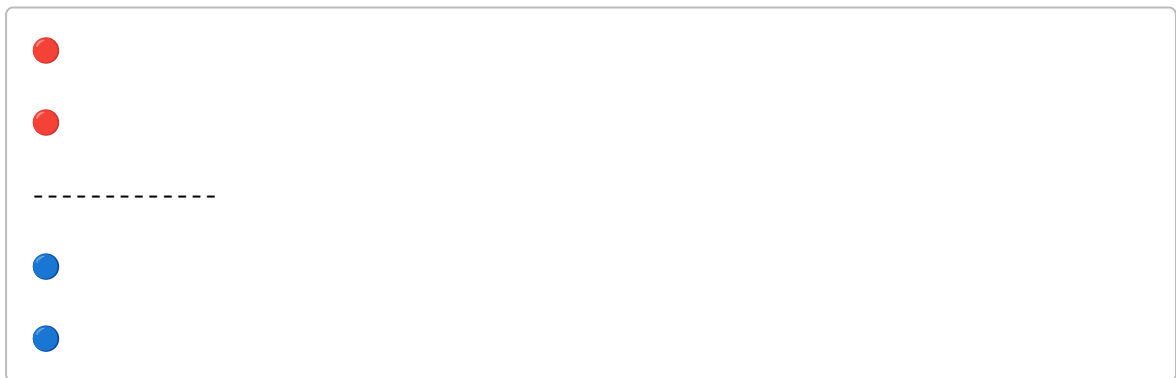
4. Support Vectors

Decision Boundary ke sabse paas wale data points ko **Support Vectors** kehte hain.

Yehi points boundary ko determine karte hain.

Door wale points ka boundary par bahut kam effect hota hai.

Example:



Boundary ke paas wale red aur blue points hi **Support Vectors** hain.

5. Hyperplane

2 Features:

$$w_1x_1 + w_2x_2 + b = 0$$

3 Features:

$$w_1x_1 + w_2x_2 + w_3x_3 + b = 0$$

General Form:

$$w \cdot x + b = 0$$

Yahan:

- **w** = Weight Vector
- **x** = Feature Vector
- **b** = Bias

6. Prediction Rule

Agar

$$w \cdot x + b > 0$$

To Prediction

Class +1

Agar

$$w \cdot x + b < 0$$

To Prediction

Class -1

Agar

$$w \cdot x + b = 0$$

To Point Decision Boundary par hai.

7. Class Labels

Logistic Regression:

0
1

SVM:

-1
+1

Ye mathematical calculations ko easy banata hai.

8. Margin Equations

Decision Boundary

$$w \cdot x + b = 0$$

Upper Margin

$$w \cdot x + b = +1$$

Lower Margin

$$w \cdot x + b = -1$$

9. Margin Formula

Maximum Margin

$$\text{Margin} = 2 / ||w||$$

Yahan

$$||w||$$

Weight Vector ki magnitude (length) hai.

Conclusion

- Small $||w|| \rightarrow$ Large Margin
 - Large Margin $\rightarrow$ Better Generalization
-

10. SVM Objective

SVM do kaam karta hai:

1. Margin ko maximize karna.
2. Classification error ko minimize karna.

11. Hinge Loss

SVM ka Loss Function

$$\text{Loss} = \max(0, 1 - y(w \cdot x + b))$$

Yahan

- y = Actual Class (+1 ya -1)
- $w \cdot x + b$ = Prediction Score

Cases

Case 1

Point margin ke bahar hai

$$\text{Loss} = 0$$

Case 2

Point margin ke andar hai

$$\text{Loss} > 0$$

Case 3

Point galat side hai

Loss bahut zyada.

12. Optimization Function

SVM minimize karta hai

$$\frac{1}{2} \|w\|^2 + C \sum (\text{Hinge Loss})$$

Isme

First Part

$$\frac{1}{2} \|w\|^2$$

Margin maximize karta hai.

Second Part

$$C \times \text{Hinge Loss}$$

Misclassification ki penalty deta hai.

13. Hard Margin

Condition

- Data perfectly separable hona chahiye.
- Kisi bhi point ki mistake allow nahi hoti.

Real life me bahut kam use hota hai.

14. Soft Margin

Condition

- Data perfectly separable nahi hota.
- Kuch mistakes allow hoti hain.

Real-world applications me mostly isi ka use hota hai.

15. Hyperparameter C

C mistake ki penalty control karta hai.

High C

- Mistakes allow nahi.
 - Complex Boundary.
 - Overfitting ka chance.
-

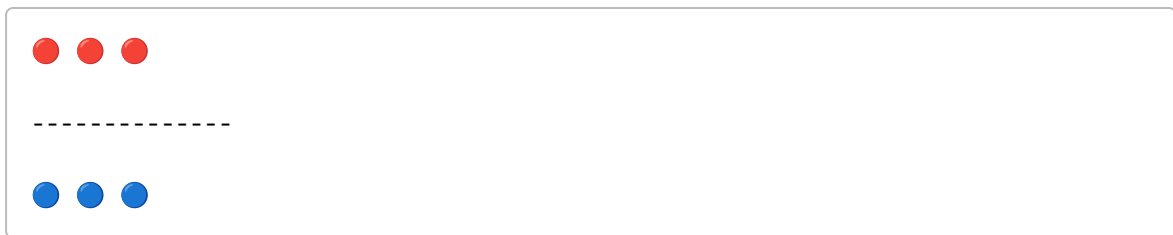
Low C

- Kuch mistakes allow.
 - Simple Boundary.
 - Better Generalization.
 - Underfitting ka chance agar bahut low ho.
-

16. Linear SVM

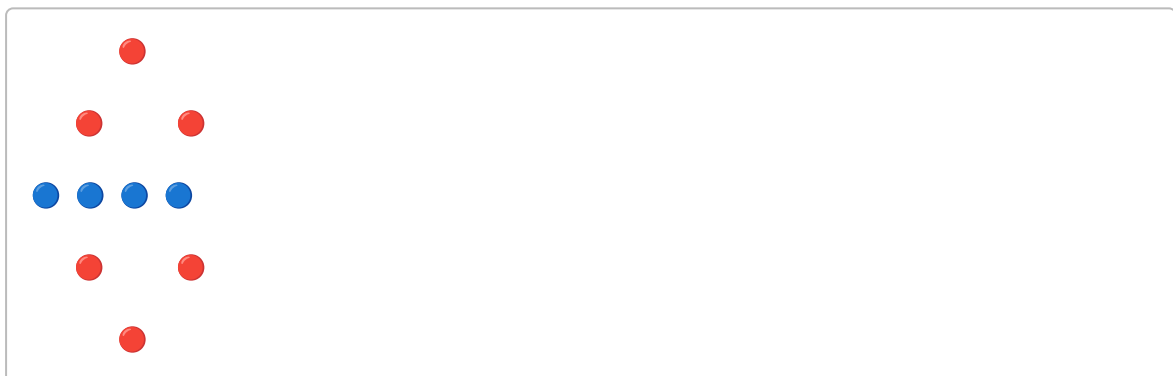
Linear SVM tab use hota hai jab data already linearly separable ho.

Example:



17. Non-Linear Data

Example



Ek straight line se separate nahi ho sakta.

18. Feature Mapping

SVM data ko higher dimensions me convert karta hai.

Example

Original Features

x
y

Mapped Features

x
y
 x^2
 y^2
xy

Higher Dimension me data linear ho sakta hai.

19. Kernel Trick

Kernel Trick bina naye features explicitly banaye higher-dimensional computation kar deta hai.

Isse

- Computation fast hoti hai.
 - Memory bachti hai.
 - High-dimensional mapping efficiently handle hoti hai.
-

20. Types of Kernels

Linear Kernel

Equation

$$K(x, z) = x^T z$$

Use:

- Linear Data

- Text Classification
 - Spam Detection
-

Polynomial Kernel

Equation

$$K(x, z) = (x^T z + 1)^d$$

d = Degree

Use:

- Moderate Non-linear Data
-

RBF (Radial Basis Function) Kernel

Equation

$$K(x, z) = e^{-\gamma ||x - z||^2}$$

Most Popular Kernel

Use:

- Complex Non-linear Data
 - Image Classification
 - Face Recognition
 - Medical Diagnosis
-

21. Gamma

Gamma decide karta hai ek training point kitni door tak influence karega.

High Gamma

- Zig-zag Boundary
 - Complex Model
 - Overfitting
-

Low Gamma

- Smooth Boundary
 - Simple Model
 - Underfitting ka chance
-

22. C vs Gamma

High C + High Gamma

- Very Complex Boundary
 - Overfitting
-

Low C + Low Gamma

- Very Simple Boundary
 - Underfitting
-

Moderate C + Moderate Gamma

Generally Best Choice

23. Important Interview Questions

Q1. SVM ka objective kya hai?

Maximum Margin Hyperplane find karna jo classification errors ko minimum rakhe.

Q2. Support Vectors kya hote hain?

Decision Boundary ke sabse paas wale data points jo boundary ko define karte hain.

Q3. Margin kya hai?

Decision Boundary aur nearest Support Vectors ke beech ki distance.

Q4. Hinge Loss kya hai?

Loss function jo margin violation aur misclassification ko penalize karta hai.

Q5. Hard Margin aur Soft Margin me difference?

Hard Margin me mistakes allow nahi hoti.

Soft Margin me kuch mistakes allow hoti hain.

Q6. Kernel Trick kya hai?

Kernel Trick bina higher-dimensional features explicitly create kiye data ko higher-dimensional space me efficiently separate karne me help karta hai.

Q7. Gamma kya control karta hai?

Decision Boundary ki complexity.

Q8. C kya control karta hai?

Misclassification ki penalty.

24. SVM Cheat Sheet

Topic	Remember
Algorithm Type	Supervised Learning
Used For	Classification & Regression
Main Goal	Maximum Margin Hyperplane
Important Points	Support Vectors
Decision Boundary	$w \cdot x + b = 0$
Margin Formula	$2/\ w\ $
Loss Function	Hinge Loss
Optimization	$\frac{1}{2}$
Hyperparameters	C, Gamma
Best Kernel	RBF
Linear Data	Linear Kernel
Non-linear Data	RBF / Polynomial Kernel

One-Line Revision

- SVM = Maximum Margin Classifier
- Support Vectors = Boundary ke nearest points
- Hyperplane = Decision Boundary
- Hinge Loss = Margin violation ko penalize karta hai
- C = Misclassification penalty
- Gamma = Boundary complexity
- Kernel Trick = Higher dimensions me efficiently mapping
- RBF = Sabse common aur powerful kernel